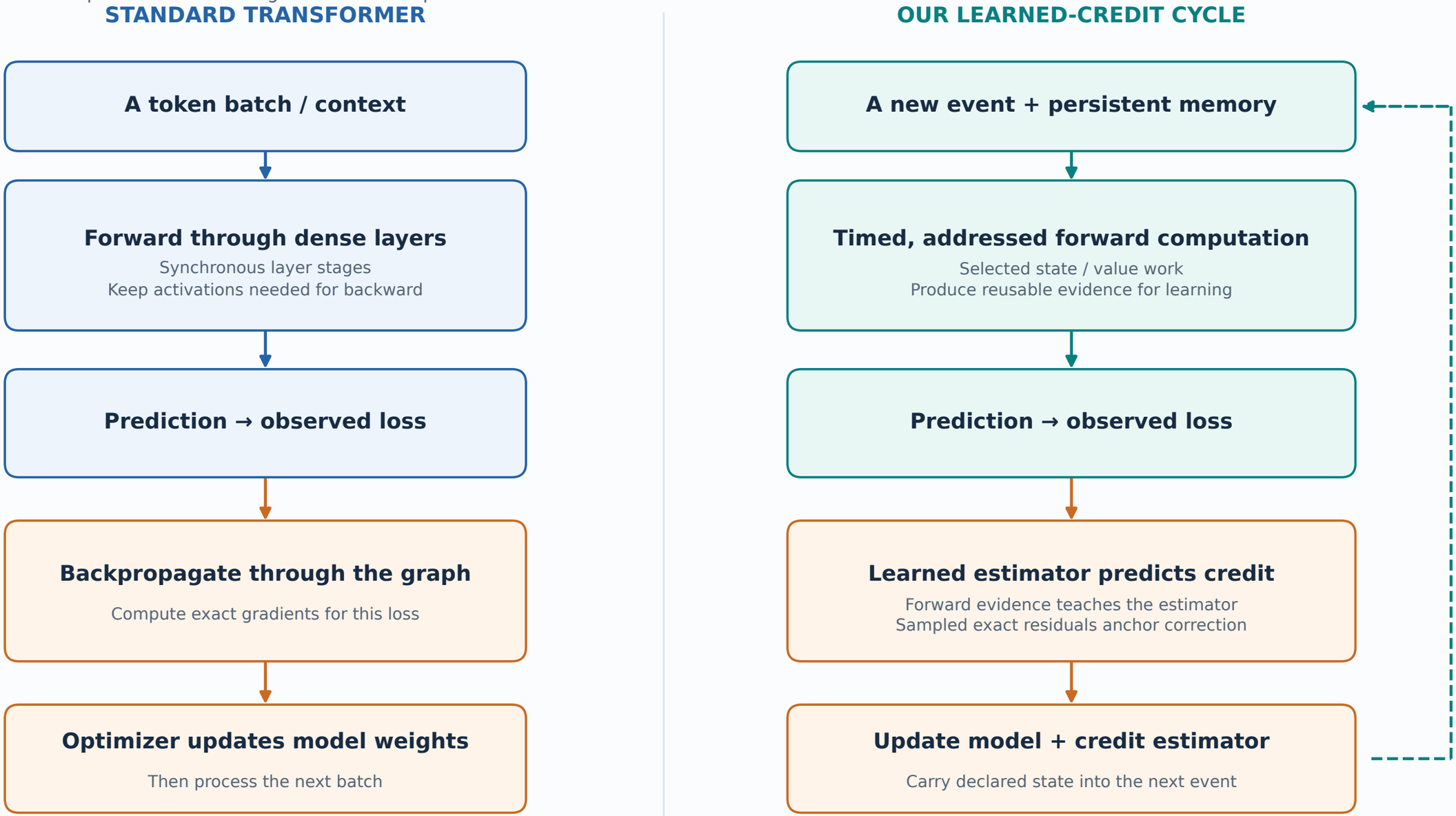
# Same goal. A different way to assign learning credit.

Ordinary synchronous Transformer training vs the proposed temporal, sparse, learned-credit substrate
Credit tells each component how to change to reduce the prediction loss.



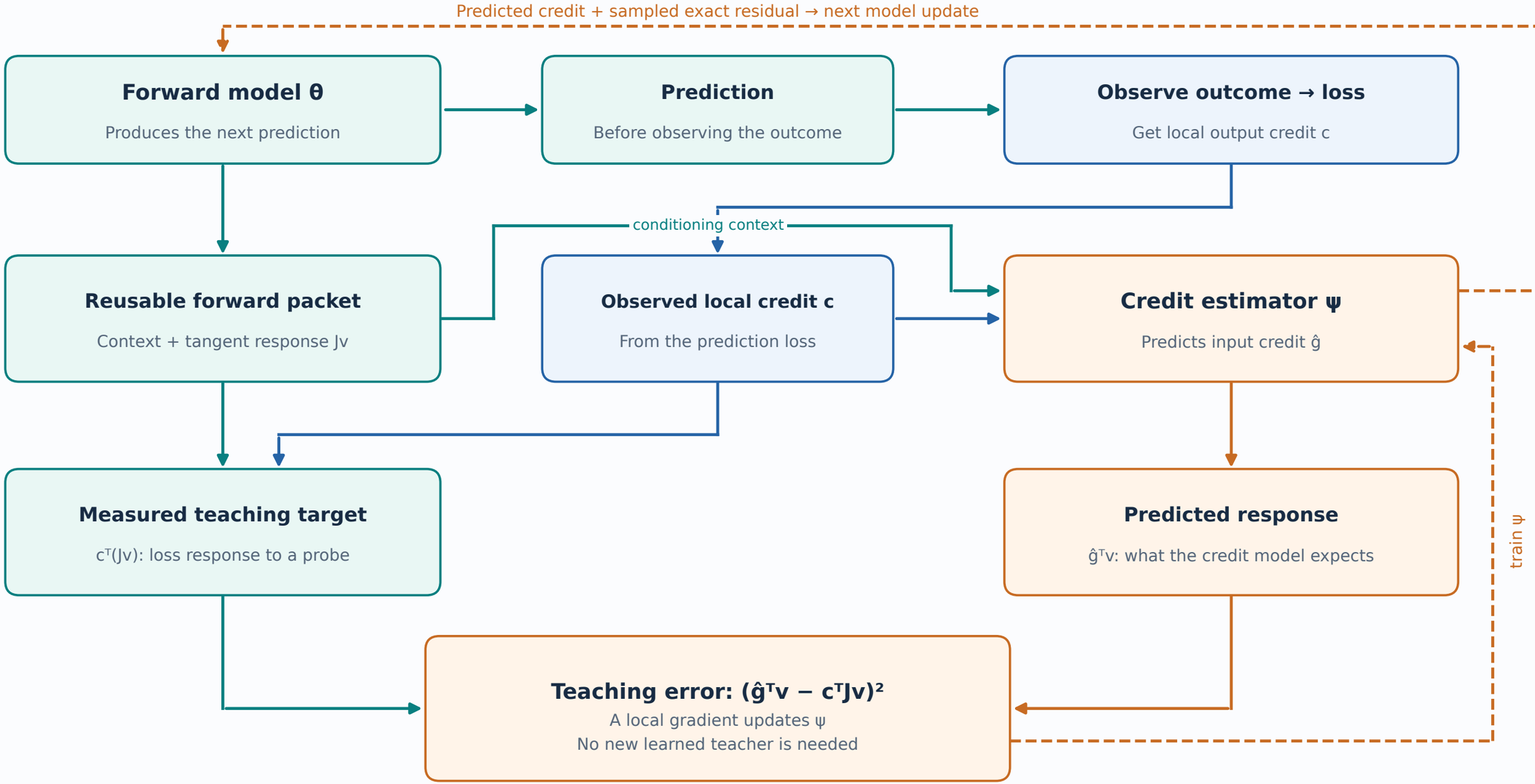

**The intended saving: reuse learned credit instead of recomputing all learning work at every event.**

Evidence scope: local/deep numerical contracts and measured component savings; the complete sparse, persistent loop is in development.

Charge discovery, counterfactuals, teaching, audits, state transport and optimizer work. No general Transformer advantage is claimed here.

# How the forward pass teaches the credit estimator

One local credit boundary; measured responses and observed outcomes close the training loop
v is an input direction; Jv its forward response; c the observed loss’s local output credit.




**Why it closes: the target is a response measured from the forward computation, anchored on an observed loss.**

Residual audit: ĝ + I/π · (g − ĝ), with positive audit probability π, preserves the conditional raw-gradient mean.

For content + time, use joint probes. Shared producer probes retain their cross terms. Versions and causal horizons must match.

This panel explains continuous credit. Hard-route interventions use future-loss advantages; their acquisition and audits are additional work.